

Minimal Model Documentation Template

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COMPANION TO *ERQI — Model Risk Testing & Analysis for Cyber Risk Quantification*

The following fields represent a minimal model documentation set for cyber risk quantification models. The level of detail should match the validation assurance level defined in *ERQI — Model Risk Testing & Analysis for Cyber Risk Quantification* — Level 1 models require comprehensive documentation across all fields; Level 3 models need at minimum the fields marked with an asterisk (*).

This template is intentionally minimal. Organizations with mature model risk management practices will supplement it with additional documentation appropriate to their context. The goal is to establish a floor, not a ceiling.

15

FIELDS

6

LEVEL-3 MINIMUM

3

ASSURANCE LEVELS

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01 Documentation Fields

FIELD	DESCRIPTION	LEVEL
Model name and version*	Identifier and version number	All
Owner*	Person or team responsible for the model	All
Validation assurance level*	Level 1, 2, or 3 based on decision criticality (see main paper §1.2)	All
Purpose and decisions supported*	What decisions does this model inform? What actions depend on its outputs?	All
Model approach	Brief description of methodology (e.g., FAIR scenario-based Monte Carlo, frequency-severity loss distribution, vendor platform)	1, 2
Key assumptions*	The 3-7 most consequential assumptions, each with a plausibility assessment and sensitivity indication	All
Known limitations*	What the model cannot do; conditions under which it should not be used; scope boundaries	All
Key inputs and data sources	Most important inputs, their sources, vintages, and known limitations. For each: empirical data (with citation), structured elicitation (with calibration notes), or analyst judgment (with rationale)	1, 2

FIELD	DESCRIPTION	LEVEL
Key outputs	What the model produces (loss exceedance curve, VaR/CVaR at specified confidence levels, sensitivity ranking, etc.)	1, 2
Uncertainty characterization	How uncertain are the outputs? At minimum: are they based on strong empirical grounding, limited data, or primarily expert judgment? For Level 1: quantified uncertainty bands	1, 2
Validation status	Summary of last validation — key findings, outstanding issues, compensating controls in place	1
Monitoring plan	What is tracked (frequency trends, severity shifts, exposure changes); what triggers revalidation	1
Last validated / Next validation due	Dates	1
Compensating controls	Conservative overlays, limits on use, minimum evidence requirements, escalation rules currently applied (see main paper §5.2)	1
Challenger model	If maintained: what alternative specification, what it showed, when last run	1

02 Guidance Notes

Classification is based on decisions, not complexity. A simple spreadsheet model used to set the board's risk appetite is Level 1. A complex model used only for internal vulnerability prioritization is Level 3. When a model informs decisions at multiple levels, classify at the highest applicable level.

Start with what you have. If a model currently has no documentation, completing the asterisked fields is a meaningful first step. Perfect documentation of a bad model is less useful than basic documentation of a good one.

Update on change. Model documentation should be updated when: the model's purpose or scope changes, a validation finding requires action, a monitoring trigger fires, or the model is recalibrated. Documentation that is created once and never revisited is a compliance artefact, not a risk management tool.

For vendor/platform models. If the model is a commercial CRQ platform, document what you can observe about the vendor's methodology and note what remains opaque. Use the companion *CRQ Vendor Validation Checklist* to assess the vendor's transparency. The model documentation template applies to your use of the platform, not to the platform itself — document your inputs, your interpretation of outputs, and your conditions of use.